

Problem-Solution Fit Document

Date	15 March 2026
Team ID	XXXXXX
Project Name	Smart Lender: AI-Based Loan Eligibility Prediction System
Maximum Marks	4 Marks

This document maps specific customer and stakeholder pain points to modular system features within the Smart Lender framework, demonstrating how each system component addresses operational bottlenecks.

S.No	Customer Pain Point (Problem)	Smart Lender Feature (Solution)	Value Created & Impact
1	Slow and labor-intensive manual vetting of thousands of loan applicants.	Automated Preprocessing & Pipeline Inference using pre-trained ML models.	Reduces approval cycles from days to under 3 seconds per application.
2	High default rates caused by human oversight and inconsistent risk assessments.	Modular Multi-Algorithm Classification Engine (Decision Trees, Random Forest, KNN, XGBoost).	Provides precise risk profiling and eliminates subjective underwriting decisions.
3	Lack of processing flexibility for institutional batch/wholesale customer reviews.	Batch CSV Upload component built directly into the web layout interface.	Allows processing of hundreds of loan applicants simultaneously in a single action.
4	Opacity in automated loan approvals causing officer hesitation and audit concerns.	Interactive Confidence Scoring with a dedicated log and dashboard history interface.	Builds immediate visual trust and provides clean records for administrative auditing.